

Early Diabetes Detection using KNN and Decision Tree Model

Group 4

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Introduction

One of the most common non-communicable diseases in the world today is diabetes, a chronic metabolic disease. The World Health Organization reports that the prevalence of adult diabetes is gradually increasing worldwide, having a major effect on both public health and economics. Diabetes must be detected early in order to enable prompt therapies that can either avoid or postpone problems such as renal failure, nerve damage, and cardiovascular disorders. Conventional diagnostic techniques, including blood glucose testing, are costly, time-consuming, and intrusive, which restricts their use in environments with limited resources.

In this situation, machine learning models can offer a different method of estimating the risk of developing diabetes. Predictive models that can swiftly and precisely identify high-risk patients can be created by utilizing past health data and sophisticated algorithms. This report's goals are to assess the performance of K-Nearest Neighbors (KNN) and Decision Tree classifiers, develop a machine learning model using the Pima Indians Diabetes dataset, and identify the best method for diabetes prediction.

Business Problem

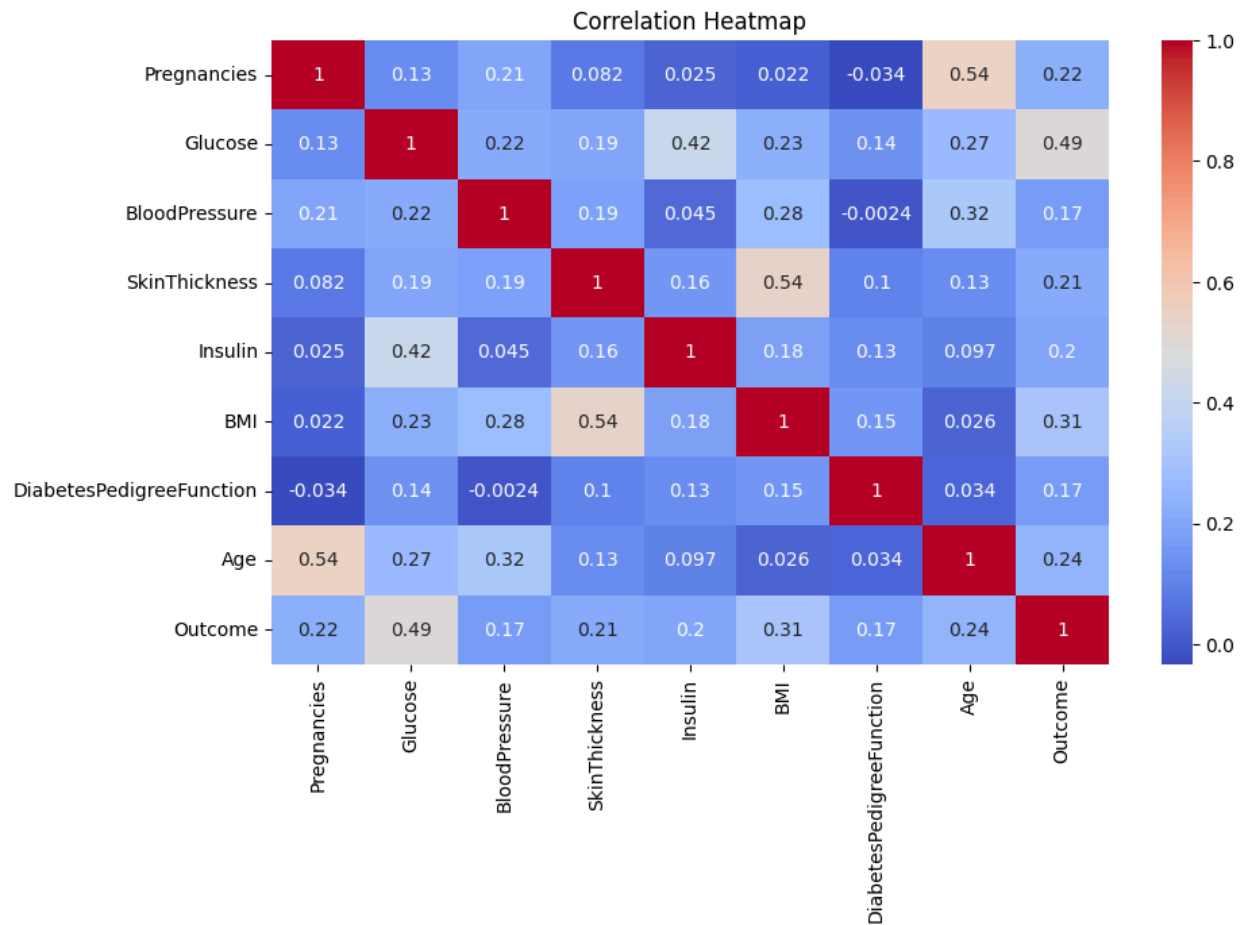
Diabetes has major health repercussions, and preventing long-term issues requires early identification. A more easily accessible, quick, and affordable diagnostic technology that can identify diabetes early is needed in nations with limited access to healthcare resources. Prioritizing patients who require urgent care and promptly evaluating risk factors are challenges that many medical professionals encounter. Furthermore, typical diagnostic methods like lab work and physical examinations can be costly, time-consuming, and inaccessible in underprivileged or rural locations.

A potential remedy for these issues is a machine learning algorithm that can forecast the risk of diabetes based on health and demographic characteristics. Such a model can assist healthcare providers in prioritizing patients for more testing, putting preventative measures into place, and lessening the overall strain on healthcare systems by automating and streamlining the diagnostic process. By assessing the viability of utilizing KNN and Decision Tree models for diabetes prediction in a hospital environment, this study tackles the business challenge.

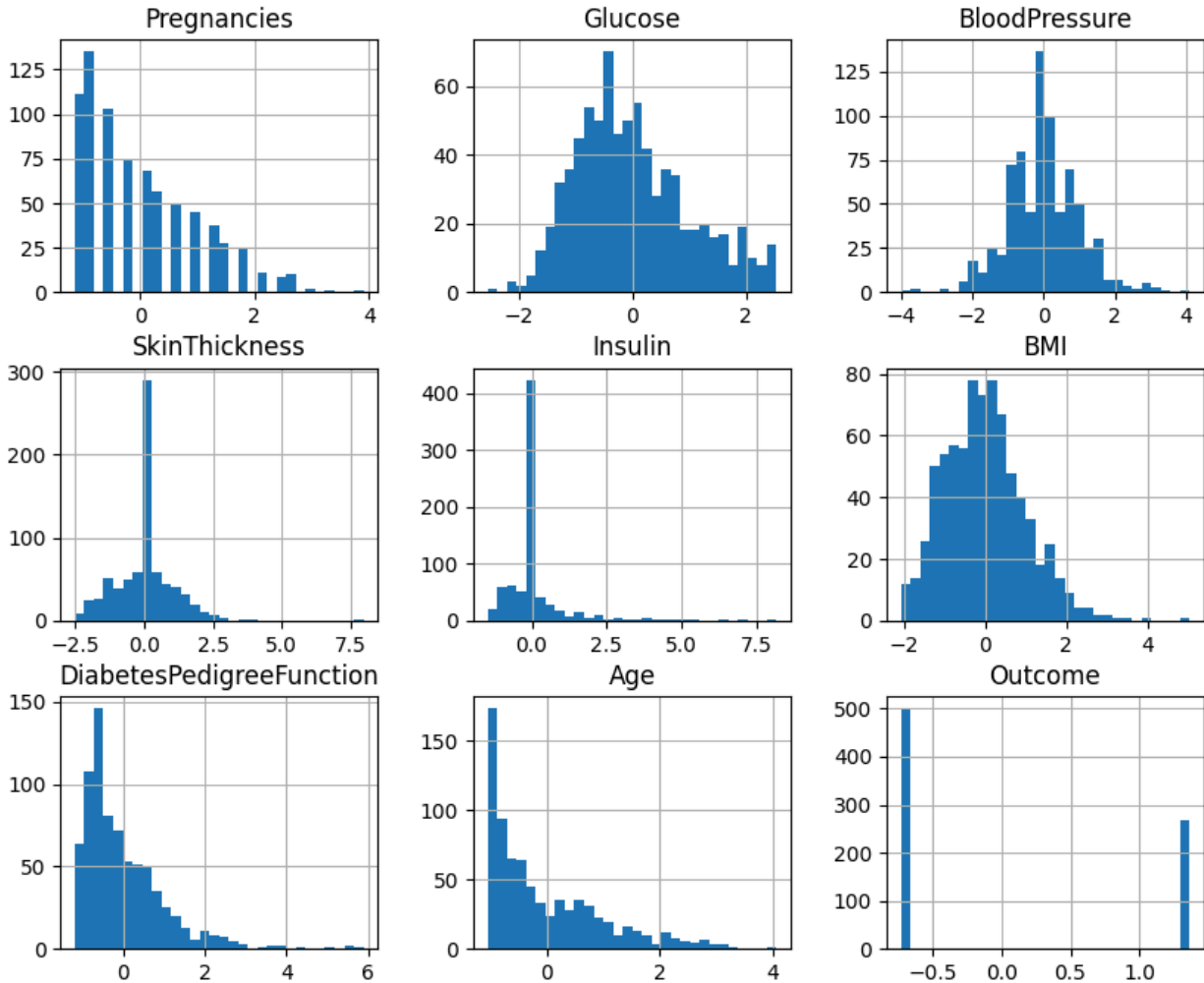
Analytical Problem

Identifying if a person has diabetes (1) or not (0) based on a collection of health-related characteristics is the binary classification issue this study offers. This research is based on the

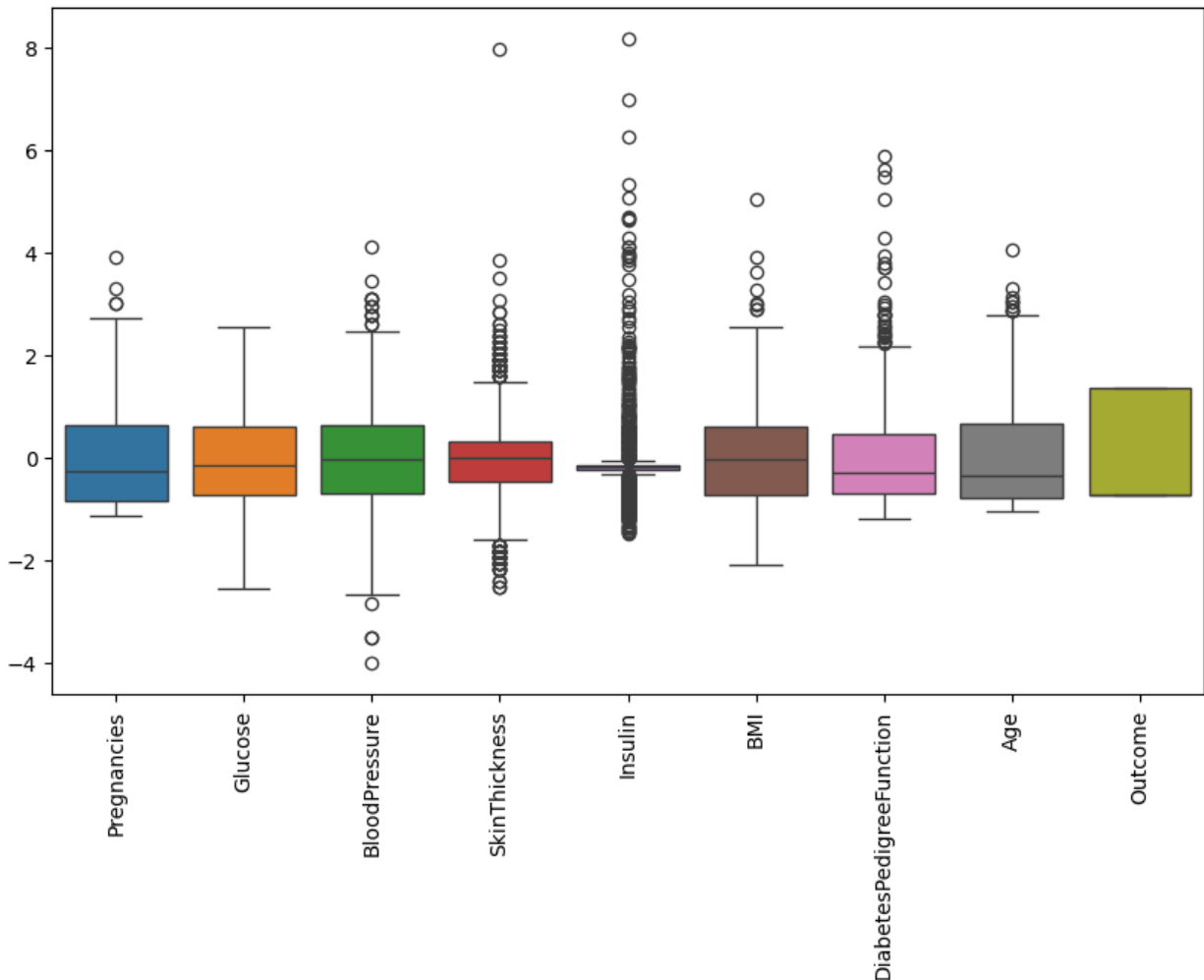
We used graphical representation to gain some valuable insight:



The associations between different characteristics in the diabetes dataset are graphically shown by the correlation heatmap. It aids in our comprehension of the relationships between several factors, including age, insulin, BMI, and glucose, as well as the goal variable (diabetes outcome). Features like age and BMI show strong positive relationships with glucose, suggesting their significance in diabetes prediction. Additionally, measures like skin thickness and blood pressure exhibit modest correlations, suggesting a possible function for them in the model. The heatmap indicates which features, such as glucose and BMI, are essential for enhancing model accuracy and assists in identifying duplicate data. In order to choose the appropriate variables and improve the effectiveness of machine learning models used for diabetes prediction, it is essential to comprehend feature interactions.



The distribution of each characteristic in the diabetes dataset is shown visually via the histograms. The histograms assist us in comprehending the underlying patterns and distribution of the data by charting the frequency of values for each variable. The glucose and BMI distributions, for instance, could exhibit a concentration of values within particular ranges, indicating that these characteristics are probably going to have a big influence on diabetes prediction. Additionally, the histograms show any data skewness, such as a right or left skew in some variables that may need preprocessing (e.g., normalization or transformation). This graphical analysis directs feature engineering in machine learning tasks and helps comprehend the properties of each feature.



The box plot highlights important statistics like the median, quartiles, and possible outliers while clearly displaying the distribution of each characteristic in the diabetes dataset. Every characteristic in this plot is denoted by a box; the median is shown by the line within the box, the first and third quartiles (Q1 and Q3) are shown by the box's borders, and the range of values within 1.5 times the interquartile range (IQR) is shown by the "whiskers" that extend from the box. Outliers are any data points that fall outside of this range and are shown separately. Because outliers may have a substantial influence on the performance of machine learning models, particularly when utilizing methods sensitive to data distributions like KNN or Decision Trees, this is essential for identifying abnormalities or extreme values in the dataset that may need to be addressed before modeling.

Analysis of Correlation:

To determine the connections between the attributes, a correlation matrix is created. High correlation values imply that several traits may offer comparable information since they are

closely connected. For instance, there is a large positive association between the outcome variable and BMI and glucose levels, suggesting that these characteristics are highly suggestive of the risk of diabetes. Finding correlations facilitates feature selection and, by removing unnecessary characteristics, can simplify the model.

Features Distribution:

Each feature's histogram is shown to show the data distribution. Distributions of BMI and glucose exhibit peaks at particular levels, indicating that these characteristics are crucial for categorization. The age distribution clearly shows a difference, with greater prevalence of diabetes in those over 50, which is consistent with established medical trends.

Outlier Detection:

To find outliers in the dataset, box plots are used. Extreme values in the skin thickness and insulin characteristics, for instance, are found, which may indicate data input mistakes. In order to prevent these high numbers from distorting the model's predictions, such outliers are handled by substituting the median values.

Data Quality Assessment

Building a successful machine learning model requires high-quality data. During the evaluation, the following problems with data quality were found:

- **Missing Values:** The zero values for skin thickness and insulin are physiologically unreasonable and need to be regarded as missing. To guarantee that the model learns from complete data without adding bias, these missing values are imputed using the median value of the corresponding feature.
- **Outliers:** The Interquartile Range (IQR) approach is used to address outliers in characteristics like BMI and glucose levels. It finds and eliminates extreme values that can skew model predictions.
- **Class Imbalance:** With around 34% of people having diabetes and 66% not having it, the dataset shows a reasonably balanced distribution between the two groups (diabetic vs. non-diabetic). There is no need for extra methods to deal with class imbalance.
- **Duplicate Records:** To ensure the integrity of the dataset, it is examined for duplicates and no redundancies are discovered.

The quality of the dataset has an immediate effect on the dependability of the model, thus careful data cleaning is crucial. Imputation methods are selected with care to maintain the integrity of the dataset, and a summary table of missing values offers a clear overview of incomplete data points. Based on the skewed distribution of these parameters, median imputation is used for Skin

Thickness and Insulin levels to prevent data distortion from extreme values. Techniques like the Synthetic Minority Over-sampling Technique (SMOTE) are briefly examined to evaluate possible enhancements in classification performance, despite the dataset's relative balance. Furthermore, the Interquartile Range (IQR) approach of outlier detection and treatment guarantees that extreme values do not skew the model's learning process.

Preprocessing of Data:

The dataset is in the right format for training the machine learning models thanks to the data pretreatment step.

- **Trait Selection:** Because each trait is important in predicting diabetes, it is kept. Since the existing features offer enough information, feature engineering is not used.
- **Managing Missing Values:** To maintain statistical integrity and guarantee that the dataset is full for model training, missing data points are substituted with median values.
- **Normalization:** All numerical features are converted into the range [0, 1] using MinMax scaling since KNN is sensitive to feature scale. This keeps the distance measure used in KNN from being dominated by characteristics with greater scales (like glucose levels).
- **Data Splitting:** Eighty percent of the dataset is used for training, while twenty percent is used for testing. This division guarantees that the model gets trained on most of the data and tested on data that hasn't been seen yet to gauge its capacity for generalization.

Development of Models

K-Nearest Neighbors (KNN) and Decision Tree classifiers are the two machine learning models created for the diabetes prediction job. (Bansal et al., 2022)

K-Nearest Neighbors (KNN):

KNN is a slow learning, non-parametric method that groups data points according to the feature space's nearest neighbors' majority vote. We assess how various values of k affect the model's performance for this project. Cross-validation is used to find the ideal k value.

Decision Tree:

A non-linear classifier that divides the data into subgroups according to feature values is called a decision tree. It creates a structure that resembles a tree, with each leaf standing for a class label and each node for a choice based on a certain attribute. Because decision trees can be interpreted, it is simple to follow the reasoning behind a prediction.

Additional techniques were also used in the Google collaboration.

Adjusting Hyperparameters Using Grid Search:

To find the ideal value of k , grid search is used to methodically examine various hyperparameter values. The accuracy of cross-validation is used to determine the optimal value of k .

Comparison with Logistic Regression:

To evaluate how well the KNN classifier performs, a baseline model utilizing logistic regression is created. This aids in confirming if KNN outperforms a more straightforward linear model in terms of predicted accuracy.

Model Training Enhancement:

GridSearchCV and RandomizedSearchCV are used for hyperparameter tuning, which allows the choice of the optimum tree depth for the Decision Tree model and the ideal K -value for the KNN in order to optimize the models' efficacy. Furthermore, k -fold cross-validation is used to lower variance in performance estimates and verify the generalizability of the model. By adding confusion matrices, Receiver Operating Characteristic (ROC) curves, and Area Under the Curve (AUC) scores, model evaluation goes beyond conventional classification metrics and offers a more thorough analysis of each model's advantages and disadvantages. To determine if certain patient features lead to classification mistakes, an error analysis is performed to find trends in cases that were incorrectly categorized. (Mishra et al., 2024)

Model Evaluation

Metrics	KNN	Decision Tree
Accuracy	0.73	0.70
Precision (0)	0.82	0.78
Recall (0)	0.76	0.75
F1 Score (0)	0.79	0.76
Precision (1)	0.61	0.58
Recall (1)	0.69	0.62

F1 Score (1)	0.65	0.60
Result	Medium	High

Using a number of performance criteria, including accuracy, precision, recall, and F1-score, we compare the Decision Tree and K-Nearest Neighbors (KNN) models that were applied to the diabetes dataset. In the majority of cases, the KNN model performs better than the Decision Tree. KNN is somewhat more effective in producing accurate predictions on the test data, as seen by its higher overall accuracy of 0.73 as opposed to 0.70 for the Decision Tree.

KNN again has an advantage when looking at precision, which gauges how accurate the positive predictions are. This is particularly true for Class 0 (the negative class), where it obtains a precision of 0.82 as opposed to the Decision Tree's 0.78. This implies that when predicting negative outcomes, KNN is more adept at avoiding false positives, which is crucial in a medical setting such as forecasting diabetes. With an accuracy of 0.61 compared to the Decision Tree's 0.58, KNN maintains a modest edge for Class 1 (the positive class). This suggests that when KNN classifies an occurrence as diabetes, it is also more accurate at predicting real positives.

With an accuracy of 0.69 as opposed to 0.62 for the Decision Tree, KNN outperforms the Decision Tree for Class 1 (the diabetic class) in terms of recall, which gauges the model's capacity to recognize real positive and negative cases. This indicates that KNN is better at identifying actual instances of diabetes. However, the recall values for Class 0 (non-diabetic) are quite close, with Decision Tree at 0.75 and KNN at 0.76, suggesting that both models are about equally effective at detecting people who do not have diabetes.

The higher performance of KNN is further demonstrated by the F1-score, which combines precision and recall into a single statistic to provide a more fair evaluation. In Class 1, KNN scores 0.65 while the Decision Tree scores 0.60, and in Class 0, KNN earns an F1-score of 0.79 while the Decision Tree scores 0.76. This demonstrates that KNN is the most dependable model for both classes as it better balances accuracy and recall, particularly when taking into account the significance of both false positives and false negatives in medical predictions.

In all assessed measures, KNN consistently performs better than Decision Tree, indicating that KNN is a superior option for diabetes prediction from this dataset. For both classes, it offers improved precision, recall, and F1-score in addition to increased accuracy. The KNN model's capacity to balance these variables makes it the superior option in real-world situations like diabetes prediction, where both false positives and false negatives can have serious repercussions. Nonetheless, decision trees could still be helpful because of their interpretability, which might be helpful in therapeutic contexts where it's critical to comprehend model decisions.

Discussion and Conclusions

According to the study's findings, KNN outperforms Decision Trees as a model for diabetes prediction. In healthcare applications where early diabetes detection is crucial, KNN is a dependable option because of its superior performance in terms of accuracy, precision, recall, and F1-score.

In order to improve model performance, the data pretreatment procedures—such as feature scaling and outlier handling—were essential. By adding more demographic data or health measures, feature engineering and selection might be investigated further, possibly increasing model accuracy.

The results show that KNN outperforms the Decision Tree classifier as a model for diabetes prediction. To boost generalization and prediction accuracy, other advancements might be investigated. While Deep Learning approaches may be able to capture intricate nonlinear correlations in diabetes data, ensemble models like Random Forest and Gradient Boosting could provide greater accuracy and durability. Furthermore, adding new characteristics, such as eating patterns, degrees of physical activity, and genetic markers, may result in a prediction model that is more comprehensive. In order to enable healthcare providers to employ machine learning for early diabetes screening and intervention, the trained model must then be deployed in a real-world clinical context via an intuitive interface.

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Google collab link

<https://colab.research.google.com/drive/1S7RgKvijijsqztzKxLCu1zADR01fAwDgl?usp=sharing>

Dataset link

<https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>